

ABHINEET MENON

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PROFESSIONAL SUMMARY

Highly adaptable Software & Data Engineer with a strong track record of designing scalable backend architectures, engineering robust data pipelines, and integrating advanced AI solutions. Passionate about solving complex technical challenges and delivering performance-driven applications across diverse technology stacks.

EDUCATION

Master of Science in Computer Science (GPA 4.0/4.0)

Aug 2024 - May 2026

University of Illinois Chicago

PROFESSIONAL EXPERIENCE

AI Intern | *BalanX Bio*

Aug 2026 - Present

- Architected a robust local real-time semantic entity extraction engine replacing legacy cloud LLM paths, effectively reducing processing latency from 1500ms to 110ms and eliminating recurring API costs.
- Engineered advanced somatic stress corroboration integrating **Genethra** wearable telemetry to dynamically modulate **Groq LLM** responses based on real-time physiological strain metrics.
- Designed and deployed a Tri-Modal Hybrid Retrieval scoring algorithm backed by **ChromaDB/Mem0**, successfully preventing context intrusion while enabling persistent user state evolution tracking.

Backend Engineer Intern | *Shop SCOUT*

Jul 2026 - Present

- Re-architected a legacy Python scraping consumer into a high-performance **asyncio** dual-lane scheduler, boosting measured throughput 2.49x and elevating automated queue drain up to 661 items per hour.
- Instrumented highly comprehensive data pipeline observability via Amazon **S3 JSONL** logs, capturing 68,000+ attempts in week one to achieve a flawless 98.3% enqueue-to-capture data reconciliation.
- Developed a robust end-of-run **C#/.NET 8** microservice securely accounting for 100% of pipeline dropped records, effectively replacing guesswork with concrete counts to diagnose untraceable data loss.

Data Engineer | *LTIMindtree*

Jun 2022 - Mar 2024

- Engineered highly robust ETL pipeline code migrating critical enterprise data from SQL Server to **Snowflake**, reliably cutting load times by 20% to significantly accelerate daily client decision-making.
- Designed and built an **Azure Data Factory** incremental-load pipeline and a companion Power BI dashboard, drastically accelerating automated data refresh cycles for key business enterprise stakeholders.
- Successfully led and mentored a 10-member engineering intern team, actively resolving complex technical blockers and facilitating daily Agile/Scrum rituals to ensure strict cross-functional project alignment.

PROJECTS

Pro Pundits League: FPL Addon Platform | *Next.js, React, Supabase, TypeScript*

- Engineered a full-stack companion platform for Fantasy Premier League featuring highly reliable real-time data sync via **GitHub Actions** cron jobs, automated scoring pipelines, and knockout tournaments.
- Built a robust scalable backend utilizing Supabase (PostgreSQL, RLS) and Next.js App Router, intelligently leveraging advanced AI workflows to dramatically accelerate server actions and database query execution.

MathSCV: Multi-Agent Reasoning Architecture | *Python, DeepSeek, Multi-Agent Systems*

- Engineered a sophisticated multi-agent debate and verification framework (Solver-Critic-Verifier) with DeepSeek-V4-Flash to effectively investigate complex mathematical reasoning and arithmetic slip correction.
- Developed a highly token-efficient evaluation engine benchmarking the SCV framework against 4 robust baselines across GSM8K, SVAMP, and ASDiv datasets to systematically quantify multi-agent failure modes.

HVAC Field Estimate Tool | *Next.js, React, Zustand, PWA, Tailwind CSS*

- Architected a robust offline-resilient Progressive Web App for field technicians to generate on-site estimates, featuring an intuitive split-screen stepper and highly advanced typo-tolerant fuzzy search.
- Engineered highly reliable persistent application state management via Zustand and dedicated CSS print media styling to seamlessly generate clean, professional PDF quotes directly on mobile client devices.

10-K Financial QA: Layout-Aware RAG | *Python, ChromaDB, DeepSeek, Streamlit*

- Architected an advanced layout-aware RAG assistant designed for zero-hallucination QA over dense SEC 10-K financial filings, meticulously validating performance via a 21-benchmark QA suite for 90.5% accuracy.
- Implemented critical spatial whitespace preservation via pdfplumber and structured character chunking to strictly maintain complex financial balance sheet alignments paired with ChromaDB semantic vector search.

TECHNICAL SKILLS

Programming: Python, C#, SQL, TypeScript, JavaScript, Java, C

Systems & Data: Next.js, React, Node.js, Snowflake, Supabase, Azure Data Factory

AI & Cloud: PyTorch, ChromaDB, RAG, AWS, Docker, Git, CI/CD